Rishi Raj, PhD

Scientific Computing | C++/Python | HPC | Data Compression

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Theoretical physicist (Sorbonne PhD, three peer-reviewed papers) building lossless C++ data compression for CMS at CERN. Strong in numerical algorithms and performance-oriented code; coursework in FPGA (VHDL, HLS), CUDA and OpenMP. Seeking engineering roles in scientific software, HPC and quantum computing.

RESEARCH EXPERIENCE

CERN | Research Placement Oct 2026 – present

CMS Next Generation Triggers | Phase-2 raw-data compression

- Designed a lossless C++ format for simulated Phase-2 Inner Tracker data: 45% smaller than the existing CMS pixel-data format with the same codec (LZMA level 4); bit-exact round trip on 400 events.
- Built reproducible codec benchmarks with a train/test split: OpenZL compressors trained per data branch gave output 13% smaller than ROOT's ZSTD level 4 at the same compression throughput.
- Evaluated lossy reduced-precision variants with full tracking on 2,000 events; technical note in preparation.

Sorbonne University (LPTHE) | Doctoral Researcher Oct 2023 – Jun 2026

- Studied black-hole microstate counting in string theory (BPS and Donaldson–Thomas invariants) using supersymmetric quantum mechanics, wall-crossing and modular forms.
- Implemented symbolic and numerical algorithms in Mathematica for BPS spectra and scattering diagrams, cross-checked against independent formulations.
- Co-authored two papers (JHEP; Annales Henri Poincaré); presented at conferences and seminars.

SELECTED PROJECTS

Personal Project | R-Distro, a Debian ARM64 Rebuild from Source 2026 – present

- Rebuilding Debian for ARM64 from a pinned snapshot as signed, immutable releases; computed the 4,725-package self-hosting set (2,732 rebuilt so far).

Research Paper | Learning Holographic Horizons (Phys. Rev. D) 2023 – 2025

- Solved the Einstein equations with pseudospectral methods for far-from-equilibrium black holes; predicted event- and apparent-horizon areas from 10 engineered time-series features (five-fold cross-validation).

Master's Thesis | C++ Simulation of the BFSS Matrix Model 2022 – 2023

- Wrote an optimised C++/Boost simulator, 100–200× faster than a baseline Mathematica implementation.
- Ran week-long campaigns of independent PBS jobs on 64+ cores to study long-time statistics and thermalisation; tracked the Gauss-law constraint and conservation of energy and angular momentum.

TECHNICAL SKILLS

Programming: C++17/20, Python (NumPy, SciPy, pandas), Mathematica, SQL, Bash.
Tools and systems: Linux, Git, CMake, Docker, pytest, ROOT, PBS batch clusters.
Numerics and ML: ODE/PDE solvers, linear algebra, optimisation, statistics; PyTorch, scikit-learn, XGBoost.
Coursework (CERN STEAM Academy): OpenMP, CUDA; VHDL, Vivado, Vitis HLS, Zynq FPGA SoCs.
Spoken languages: English (fluent), French (A1), German (A1).

EDUCATION AND TRAINING

PhD in Theoretical Physics | Sorbonne University 2023 – 2026

Thesis: *BPS Black Holes and Donaldson–Thomas Invariants* (defended Jun 2026).

CERN STEAM Academy | Scientific Computing, Hardware and ML Jun – Aug 2026

Ten-week full-time programme; grade 5.5/6.

BS (Hons) and MS in Theoretical Physics | IIT Madras 2018 – 2023

Minor in Mathematics. MS GPA 9.59/10; overall GPA 9.18/10.

AWARDS AND ACTIVITIES

Awards: COST Short-Term Scientific Mission grant (Geneva, 2026); Charpak Lab Fellowship (France, 2023); 63rd and 64th Institute Day Prizes (IIT Madras); KVPY and NIUS fellowships (India, 2018–2023).
Activities: CERN Patatrack Hackathon (2026), ROOT Fixathon (2024); 4EU+ ML-in-Physics PhD School (2025).